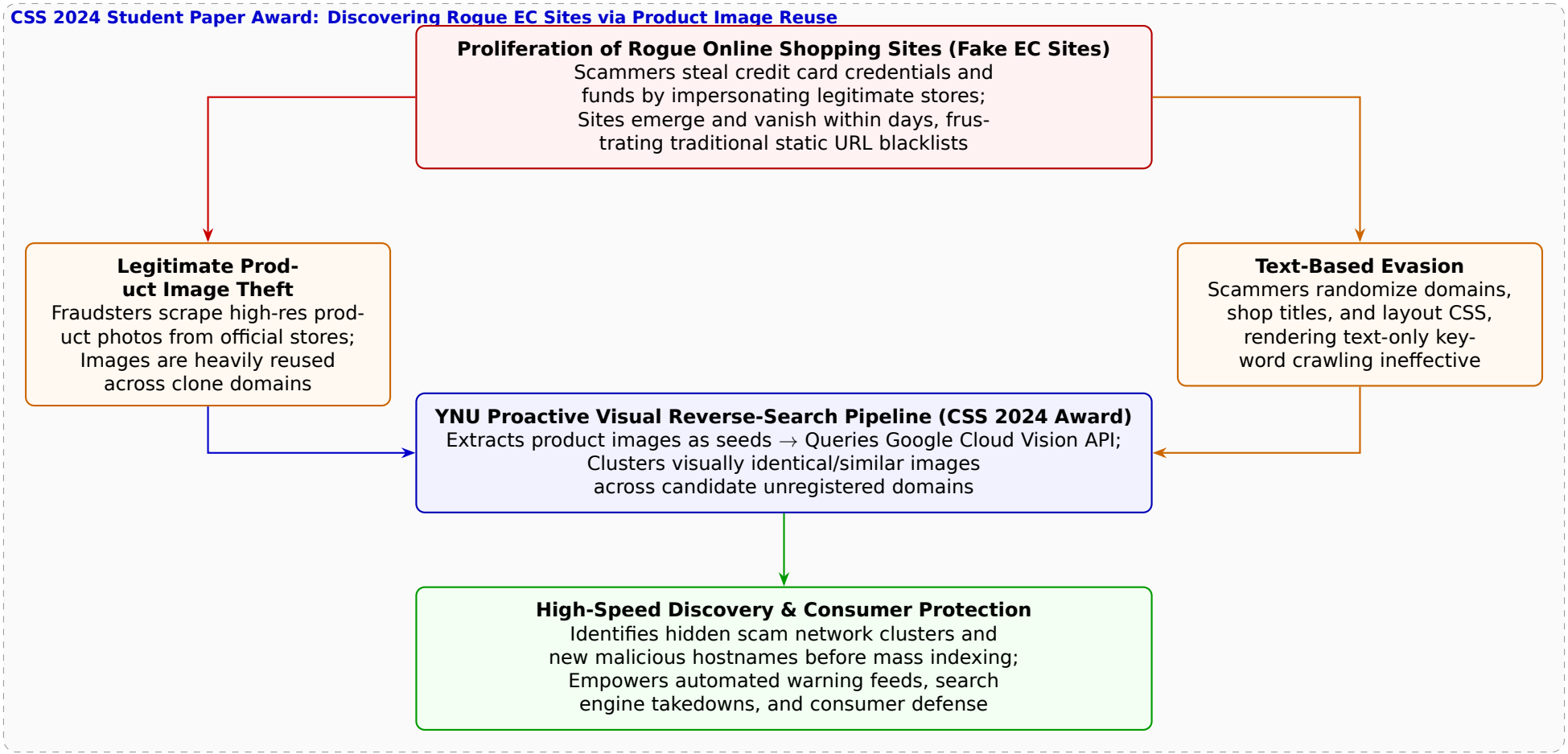
CSS 2024 Student Paper Award: Discovering Rogue EC Sites via Product Image Reuse
Proliferation of Rogue Online Shopping Sites (Fake EC Sites)
Scammers steal credit card credentials and funds by impersonating legitimate stores; Sites emerge and vanish within days, frustrating traditional static URL blacklists
Legitimate Product Image Theft
Fraudsters scrape high-res product photos from official stores; Images are heavily reused across clone domains
Text-Based Evasion
Scammers randomize domains, shop titles, and layout CSS, rendering text-only keyword crawling ineffective
YNU Proactive Visual Reverse-Search Pipeline (CSS 2024 Award)
Extracts product images as seeds → Queries Google Cloud Vision API; Clusters visually identical/similar images across candidate unregistered domains
High-Speed Discovery & Consumer Protection
Identifies hidden scam network clusters and new malicious hostnames before mass indexing; Empowers automated warning feeds, search engine takedowns, and consumer defense